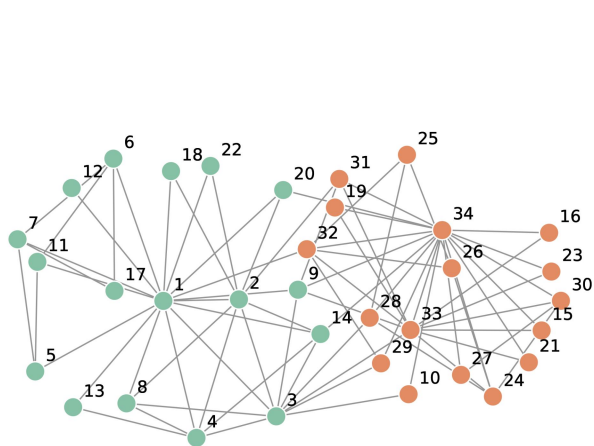
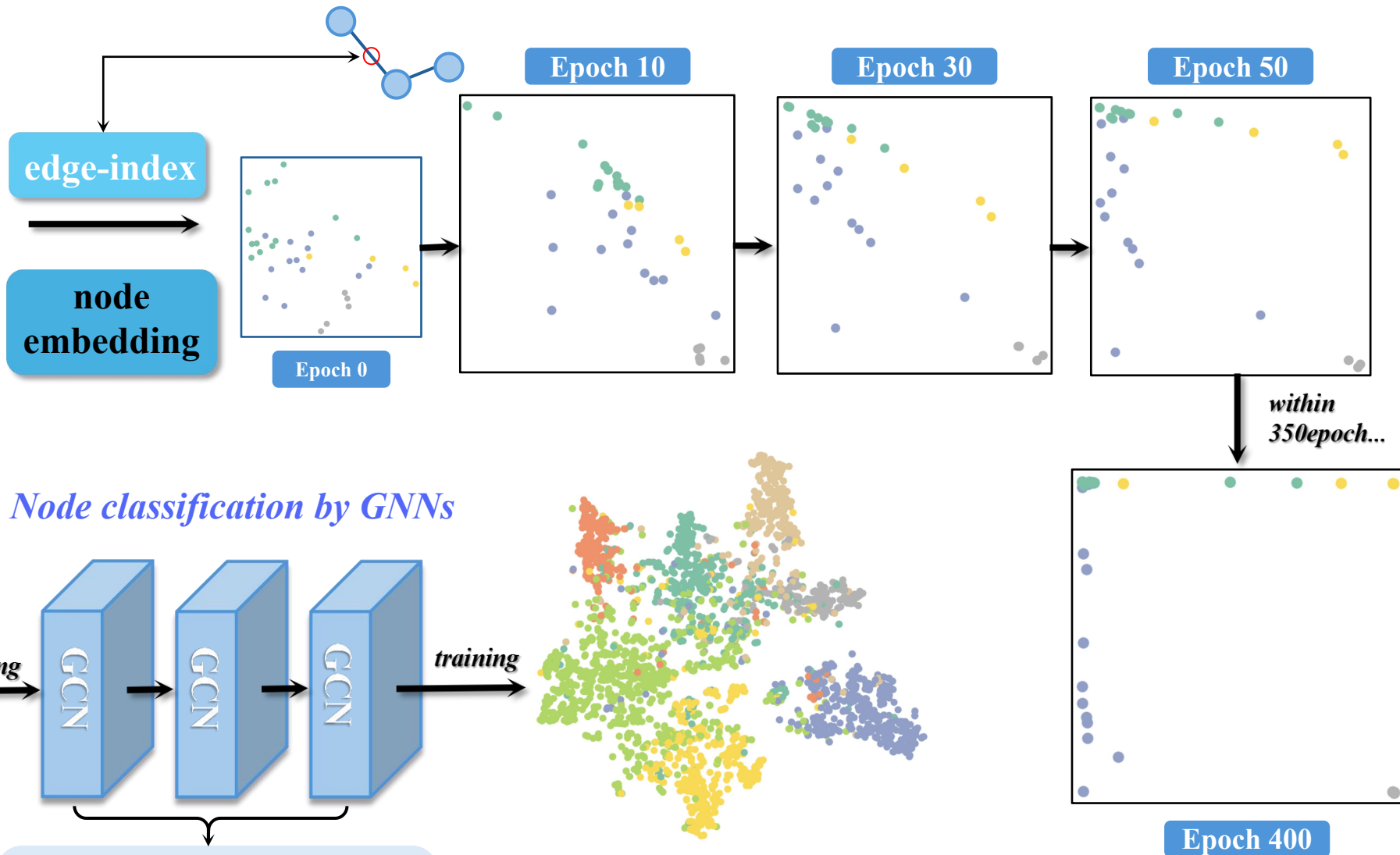
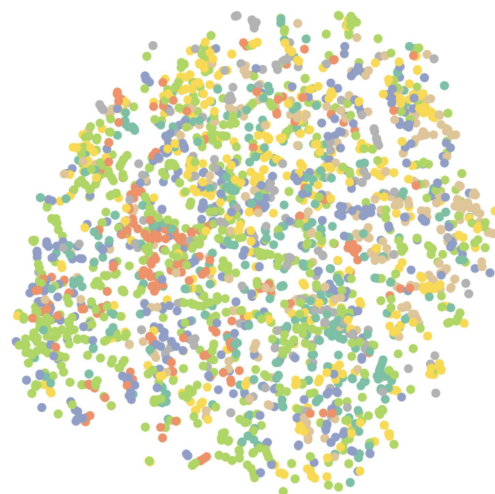




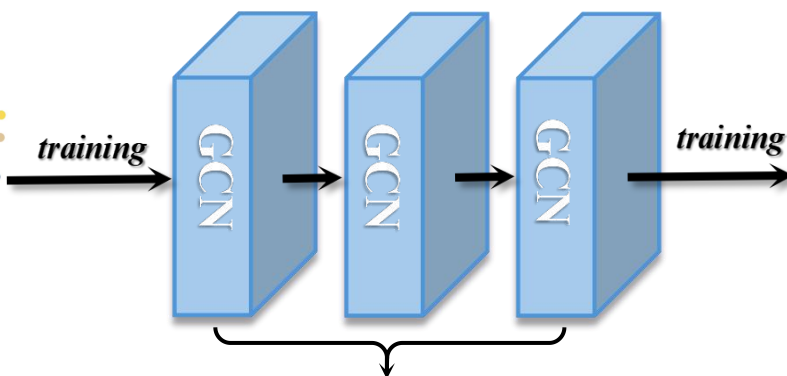
Karate Club Network



Node classification by GNNs



Initialized nodes



nodes after classification

Graph Convolutional layer

$$\mathbf{x}_v^{(\ell+1)} = \mathbf{W}^{(\ell+1)} \sum_{w \in \mathcal{N}(v) \cup \{v\}} \frac{1}{c_{w,v}} \cdot \mathbf{x}_w^{(\ell)}$$

$$\begin{matrix} \mathcal{G}_1 = (\mathbf{X}_1, \mathbf{A}_1) \\ \mathcal{G}_2 = (\mathbf{X}_2, \mathbf{A}_2) \end{matrix} \begin{matrix} \text{graph} \\ \text{graph} \end{matrix} \Rightarrow \text{GNN} \left(\begin{bmatrix} \mathbf{A}_1 & \\ & \mathbf{A}_2 \end{bmatrix}, \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{X}'_1 \\ \mathbf{X}'_2 \end{bmatrix}$$